THE FINITE FREE STAM INEQUALITY

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Let $\boxplus_n$ and $\Phi_n(\cdot)$ be defined as in the problem statement. In this note we prove the following result, which was conjectured by D. Shlyakhtenko.

Theorem 0.1. *Let $p(x)$ and $q(x)$ be any two monic real-rooted polynomials of degree n. Then*

$$\frac{1}{\Phi_n(p \boxplus_n q)} \geq \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}.$$

1. Notation and preliminaries

1.1. **Polynomials and the finite free convolution.** Given a polynomial $p(x)$ of degree n we say that $\alpha = (\alpha_1, \ldots, \alpha_n)$ is a vector of roots for $p(x)$ if the α_i are the roots of $p(x)$. We will say that α is ordered if $\alpha_1 \geq \cdots \geq \alpha_n$. Recall that for monic polynomials $p(x)$ and $q(x)$, $p(x) \boxplus_n q(x)$ may be expressed as:

$$p(x) \boxplus_n q(x) = \sum_{\pi \in S_n} \prod_{i=1}^{n} (x - \alpha_i - \beta_{\pi(i)}), \tag{1.1}$$

where α and β are vectors of roots for $p(x)$ and $q(x)$, respectively, and S_n is the symmetric group on n elements (see Theorem 2.11 of [**?**] for a proof). Walsh [**?**] proved that if $p(x)$ and $q(x)$ are real-rooted, then so is $p(x) \boxplus_n q(x)$. Therefore, the finite free convolution induces a map

$$\Omega_{\boxplus_n} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n,$$

where if α and β are vectors of roots for $p(x)$ and $q(x)$, then $\Omega_{\boxplus_n}(\alpha, \beta)$ is defined to be the ordered vector of roots for $p(x) \boxplus_n q(x)$.

Other than the fact that $\boxplus_n$ preserves real-rootedness, our proof will crucially exploit each of the following well-known properties of the finite free convolution. In what follows we will use $\mathbb{1}_n$ to denote the all-ones vector of dimension n. We will use the notation

$$m_k(\alpha) := \frac{1}{n} \sum_{i=1}^{n} \alpha_i^k \quad \text{and} \quad \mathrm{Var}(\alpha) := m_2(\alpha) - m_1(\alpha)^2.$$

Proposition 1.1 (Properties of $\boxplus_n$)**.** *If $\alpha, \beta \in \mathbb{R}^n$ and $\gamma = \Omega_{\boxplus_n}(\alpha, \beta)$, then:*

i) (Additivity) $m_1(\gamma) = m_1(\alpha) + m_1(\beta)$ and $\mathrm{Var}(\gamma) = \mathrm{Var}(\alpha) + \mathrm{Var}(\beta)$.

ii) (Commutation with translation) For all $t \in \mathbb{R}$, $\Omega_{\boxplus_n}(\alpha + t\mathbb{1}_n, \beta) = \gamma + t\mathbb{1}_n$ and $\Omega_{\boxplus_n}(\alpha, \beta + t\mathbb{1}_n) = \gamma + t\mathbb{1}_n$.

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Proof. (i) Follows from the definition of $p \boxplus_n q$ in terms of the coefficients of p and q and the Newton identities. (ii) Follows from (**??**). $\square$

1.2. **The heat flow and the finite free Fisher information.** Given a vector of roots $\alpha \in \mathbb{R}^n$ we will define the its finite free score vector $\mathscr{J}_n(\alpha) \in (\mathbb{R} \cup \{\infty\})^n$ as

$$\mathscr{J}_n(\alpha) := \left(\sum_{j:j \neq i} \frac{1}{\alpha_i - \alpha_j} \right)_{i=1}^n .$$

Given a real-rooted polynomial $p(x)$ with vector of roots α, define its finite free Fisher information as

$$\Phi_n(p) := \| \mathscr{J}_n(\alpha) \|^2.$$

The following fact will allow us to write the finite free Fisher information of the polynomial $p(x)$ in terms of the dynamics of its roots under the reverse heat flow. It was shown to us by D. Shlyakhtenko.

Lemma 1.1 (Score vectors as derivatives)**.** *Assume $p(x)$ has simple roots. Let $p_t(x) := \exp\left(-\frac{t}{2}\partial_x^2\right) p(x)$ and let $\alpha(t) = (\alpha_1(t), \ldots, \alpha_n(t))$ be the ordered vector of roots of $p_t(x)$. Then*

$$\alpha_i'(0) = \sum_{j:j \neq i} \frac{1}{\alpha_i - \alpha_j},$$

and in particular $\alpha'(0) = \mathscr{J}_n(\alpha)$.

Proof. Since the $\alpha_i(t)$ are continuous in t, the roots remain simple in a neighborhood of $t = 0$. Implicitly differentiating the expression

$$p(\alpha_i(t)) - tp''(\alpha_i(t))/2 + t^2 R(\alpha_i(t), t) = 0$$

(where $R(x, t)$ is a polynomial) at $t = 0$ one obtains

$$\alpha_i'(0) = \frac{1}{2} \frac{p''(\alpha_i)}{p'(\alpha_i)},$$

which is equal to the advertised expression. $\square$

2. Proof of Stam's inequality

We now prove Theorem **??**. The following Lemma allows us to restrict attention to the case when p, q, and $p \boxplus_n q$ all have simple roots.

Lemma 2.1 (Approximation by Simple Rooted Polynomials)**.** *Let $\epsilon > 0$ and define the differential operator $T_\epsilon := (1 - \epsilon \cdot d/dx)^n$. If $p(x)$ is a monic real-rooted polynomial of degree n, then*

i) $(T_\epsilon p)(x)$ is monic and real-rooted of degree n with simple roots.
ii) $\Phi_n(T_\epsilon p) \to \Phi_n(p)$ as $\epsilon \to 0$.
iii) $(T_\epsilon p) \boxplus_n (T_\epsilon q) = T_\epsilon^2 (p \boxplus_n q)$.

Proof. (i) was shown in [**?**]. (ii) is because Φ_n is continuous in the roots of p, which are continuous in ϵ. (iii) follows because $\boxplus_n$ commutes with differential operators (see e.g. [**?**].) $\square$

Thus, establishing Theorem **??** for the simple case implies the general case by using (iii) above and taking $\epsilon \to 0$. In what follows, $p(x)$ and $q(x)$ are monic real-rooted polynomials, α and β are vectors of roots for $p(x)$ and $q(x)$, $\gamma := \Omega_{\boxplus_n}(\alpha, \beta)$, and α, β, γ all have distinct entries, implying that they are smooth functions of the coefficients of the corresponding polynomials. Let $J_{\boxplus_n}$ denote the Jacobian of $\Omega_{\boxplus_n}$ at the point (α, β).

Our proof can be separated into three steps. The second step is the most substantial one and we will defer its detailed discussion to Section **??**.

Step 1 (Jacobians and score vectors). We first note that the following relation between score vectors holds.

Observation 2.1 (Relating score vectors)**.** *Using the above notation, for any $a, b \geq 0$*

$$J_{\boxplus_n}(a\,\mathscr{J}_n(\alpha), b\,\mathscr{J}_n(\beta)) = (a+b)\,\mathscr{J}_n(\gamma).$$

Proof. For every $t \geq 0$ let $p_t(x) = \exp(-\frac{t}{2}\partial_x^2)p(x)$, let $\alpha(t)$ be the ordered vector of roots of p_t, and define q_t, r_t and $\beta(t), \gamma(t)$ in an analogous way. Since the finite free convolution commutes with any differential operator, it follows that

$$r_{(a+b)t} = p_{at} \boxplus_n q_{bt}.$$

Hence $\gamma((a+b)t) = \Omega_{\boxplus_n}(\alpha_{at}, \beta_{bt})$ for every t. So, if we differentiate this relation with respect to t, using the chain rule for the right-hand side, we get

$$(a+b)\gamma'(0) = J_{\boxplus_n}\begin{pmatrix} a \cdot \alpha'(0) \\ b \cdot \beta'(0) \end{pmatrix}.$$

A direct application of Lemma **??** concludes the proof. $\square$

Step 2 (Understanding the Jacobian). The substance of our proof lies in understanding $J_{\boxplus_n}$. In particular, we will show the following.

Proposition 2.1. *If $u, v \in \mathbb{R}^n$ are orthogonal to $\mathbb{1}_n$ then*

$$\|J_{\boxplus_n}(u, v)\|^2 \leq \|u\|^2 + \|v\|^2.$$

This proposition will be proven in Section **??**, for now we show how it is used.

Step 3 (Proof of Theorem ?? à la Blachman). With Observation **??** and Proposition **??** in hand we can conclude the proof using the same argument that Blachman used in [**?**].

Proof of Theorem **??**. First note that

$$\sum_{i=1}^{n} \sum_{j:j\neq i} \frac{1}{\alpha_i - \alpha_j} = 0,$$

since each term in the sum appears once with a plus and once with a minus. Therefore $\mathscr{J}_n(\alpha)$ is orthogonal to $\mathbb{1}_n$ and, arguing analogously, $\mathscr{J}_n(\beta)$ is orthogonal to $\mathbb{1}_n$. So, Proposition **??** implies

$$\|J_{\boxplus_n}(a\mathscr{J}_n(\alpha), b\mathscr{J}_n(\beta))\|^2 \leq a^2\|\mathscr{J}_n(\alpha)\|^2 + b^2\|\mathscr{J}_n(\beta)\|^2.$$

Combining this with Observation **??** yields

$$(a+b)^2\|\mathscr{J}_n(\gamma)\|^2 \leq a^2\|\mathscr{J}_n(\alpha)\|^2 + b^2\|\mathscr{J}_n(\beta)\|^2.$$

Now, by choosing $a = \frac{1}{\|\mathscr{J}_n(\alpha)\|^2}$ and $b = \frac{1}{\|\mathscr{J}_n(\beta)\|^2}$, the above inequality turns into

$$\left(\frac{1}{\|\mathscr{J}_n(\alpha)\|^2} + \frac{1}{\|\mathscr{J}_n(\beta)\|^2}\right)^2 \|\mathscr{J}_n(\gamma)\|^2 \leq \frac{1}{\|\mathscr{J}_n(\alpha)\|^2} + \frac{1}{\|\mathscr{J}_n(\beta)\|^2},$$

which after simple algebraic manipulations can be turned into the inequality claimed in Theorem **??**. □

2.1. **Understanding $J_{\boxplus_n}$.** Let $(\Omega_{\boxplus_n,1}, \ldots, \Omega_{\boxplus_n,n})$ be the coordinate functions of $\Omega_{\boxplus_n}$, that is $\gamma_i = \Omega_{\boxplus_n,i}(\alpha,\beta)$. The starting point of our approach to proving Proposition **??** is the observation that the matrix $J_{\boxplus_n}J^*_{\boxplus_n}$ is related to the Hessians of the functions $\Omega_{\boxplus_n,i}$. It will be helpful to introduce the notation

$$H^{(i)}_{\boxplus_n} := \mathrm{Hess}_{\Omega_{\boxplus_n,i}}.$$

For this discussion it will prove useful to define the $(2n-2)$-dimensional subspace

$$\mathcal{V} = \{(u,v) \in \mathbb{R}^n \times \mathbb{R}^n : u^*\mathbb{1}_n = v^*\mathbb{1}_n = 0\}.$$

And, given $w \in \mathbb{R}^n \times \mathbb{R}^n$ and $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ we will use $D_w f$ to denote the directional derivative of f in the direction of w, that is $D_w = \sum_i w_i \partial_i$.

Lemma 2.2 (The Hessian of $\Omega_{\boxplus_n}$)**.** *Using the above notation*

$$w^* J_{\boxplus_n} J^*_{\boxplus_n} w = w^*\Big(I_n \oplus I_n - \sum_{i=1}^n \gamma_i H^{(i)}_{\boxplus_n}\Big)w, \quad \forall w \in \mathcal{V}. \tag{2.1}$$

Proof. Fix $w = (u,v) \in \mathcal{V}$ and define

$$\alpha(t) := \alpha + tu, \quad \beta(t) := \beta + tv, \quad \text{and} \quad \gamma(t) := \Omega_{\boxplus_n}(\alpha(t), \beta(t)),$$

and note that the variance additivity from Proposition **?? ??**) implies that

$$m_2(\gamma(t)) - m_1(\gamma(t))^2 = m_2(\alpha(t)) + m_2(\beta(t)) - (m_1(\alpha(t))^2 + m_1(\beta(t))^2).$$

Now, the fact that $(u,v) \in \mathcal{V}$ implies that the means $m_1(\alpha(t))$ and $m_1(\beta(t))$ are a constant function of t and therefore, again by Proposition **?? ??**), the mean $m_1(\gamma(t))$ is also a constant function of t. So, differentiating the above equation twice with respect to t we get

$$\partial_t^2 m_2(\gamma(t))\big|_{t=0} = \partial_t^2\big(m_2(\alpha(t)) + m_2(\beta(t))\big)\big|_{t=0}. \tag{2.2}$$

Now we inspect both sides of the above equation. First

$$n\,\partial_t^2 m_2(\gamma(t))\big|_{t=0} = \sum_{i=1}^n D_w^2(\gamma_i^2)$$

$$= 2\sum_{i=1}^{n}\left((D_w\gamma_i)^2 + \gamma_i D_w^2\gamma_i\right)$$

$$= 2\left(w^* J_{\boxplus_n} J_{\boxplus_n}^* w + \sum_{i=1}^{n}\gamma_i w^* H_{\boxplus_n}^{(i)} w\right). \tag{2.3}$$

Second

$$\begin{aligned} n\,\partial_t^2(m_2(\alpha(t)) + m_2(\beta(t))) &= \partial_t^2((\alpha+tu)^*(\alpha+tu) + (\beta+tv)^*(\beta+tv)) \\ &= 2(u^*u + v^*v) \\ &= 2w^*w. \end{aligned} \tag{2.4}$$

Finally, plugging (**??**) and (**??**) back into (**??**) yields

$$w^* J_{\boxplus_n} J_{\boxplus_n}^* w + \sum_{i=1}^{n}\gamma_i w^* H_{\boxplus_n}^{(i)} w = w^*w,$$

which is equivalent to the advertised result. □

We now apply a result of Bauschke et al. [**?**, Corollary 3.3].

Theorem 2.2 (Bauschke et al.)**.** *Let $f \in \mathbb{R}[x_1,\dots,x_m]$ be a hyperbolic polynomial in the direction $w \in \mathbb{R}^m$ and for every $a \in \mathbb{R}^m$ let $\lambda_1(a) \geq \cdots \geq \lambda_m(a)$ be the roots of $g_a(t) := f(a+tw)$. Then, for every $k = 1,\dots,m$, the function $\sigma_k(a) := \sum_{i=1}^k \lambda_i(a)$ is convex in a.*

In our context this implies the following.

Corollary 2.1. *For any real numbers $c_1 \geq \cdots \geq c_n$, the matrix $\sum_{i=1}^n c_i H_{\boxplus_n}^{(i)}$ is PSD.*

Proof. Define the multivariate polynomial

$$f(x, a_1,\dots,a_n,b_1,\dots,b_n) := \sum_{\pi\in S_n}\prod_{i=1}^{n}(x - a_i - b_{\pi(i)}).$$

Since the above polynomial is homogeneous and the finite free convolution preserves real rootedness, f is hyperbolic in the direction $e_1 = (1, 0\cdots, 0)$. Now, by Theorem **??** the functions

$$\sigma_k(x,a,b) = \sum_{i=1}^{k}\lambda_i(x,a,b)$$

are convex, where $\lambda_1(x,a,b) \geq \cdots \geq \lambda_n(x,a,b)$ denote the roots of $f((x,a,b)+te_1)$. And, because the c_i are ordered we moreover have that the function

$$L(x,a,b) := \sum_{i=1}^{n} c_i\lambda_i(x,a,b)$$

is convex, as it can be written as a positive linear combination of the σ_k. It follows that $\mathrm{Hess}_L = \sum_{i=1}^n c_i \mathrm{Hess}_{\lambda_i}$ at any (x,a,b) is PSD. But, on the other hand, when $x = 0$, $a = \alpha$ and $b = \beta$, we have that $\mathrm{Hess}_{\lambda_i} = H_{\boxplus_n}^{(i)}$, which in turn gives that $\sum_{i=1}^n c_i H_{\boxplus_n}^{(i)}$ is PSD. □

We can now complete the proof of Proposition **??**.

Proof of Proposition **??**. Let $(u, v) \in \mathcal{V}$. Then

$$\|J_{\boxplus_n}(u,v)\|^2 = (u,v)^* J_{\boxplus_n} J^*_{\boxplus_n}(u,v) = \|u\|^2 + \|v\|^2 - \sum_{i=1}^{n} \gamma_i (u,v)^* H^{(i)}_{\boxplus_n}(u,v),$$

where the last equality follows from Lemma **??**. Now, applying Corollary **??** with $c_i = \gamma_i$ gives that $\sum_{i=1}^{n} \gamma_i H^{(i)}_{\boxplus_n}$ is PSD, and hence

$$\sum_{i=1}^{n} \gamma_i (u,v)^* H^{(i)}_{\boxplus_n}(u,v) \geq 0.$$

The proof follows from putting the two expressions together. $\square$